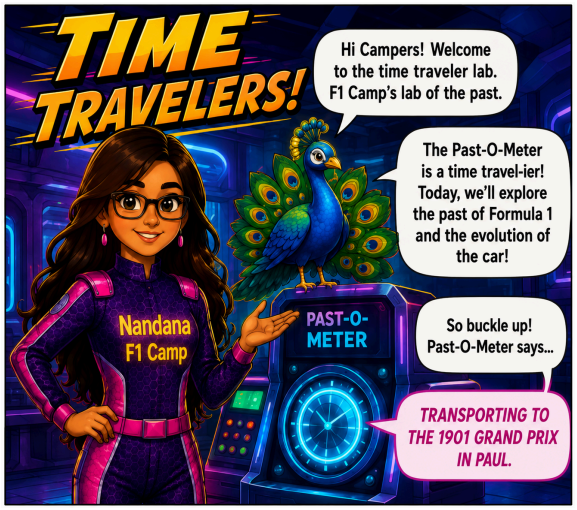
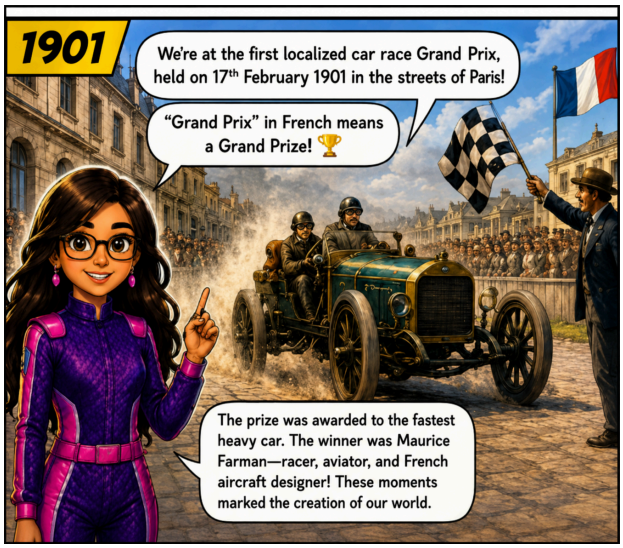
TIME TRAVELLERS

Hi Campers! Welcome to the time traveller lab. Formula Racing Camp's lab of the past. The lab has a very cool machine, the Past-O-Meter. The machine is a time traveller. Isn't that cool? Today, this machine is going to help us to explore the past of Formula. Today we will explore the evolution of the car. So take your seats in the Past-O-Meter, so we can go to the start of F1 racing. **Past-O-Meter says, "Transporting to the 1901 Grand Prix in Pau."**



1901



So here we are, at the first localised car race Grand Prix held before F1 had begun on 17th February 1901 in the streets of Pau. **Did you know the Grand Prix in French means a Grand Prize?** In this race, the prize was awarded

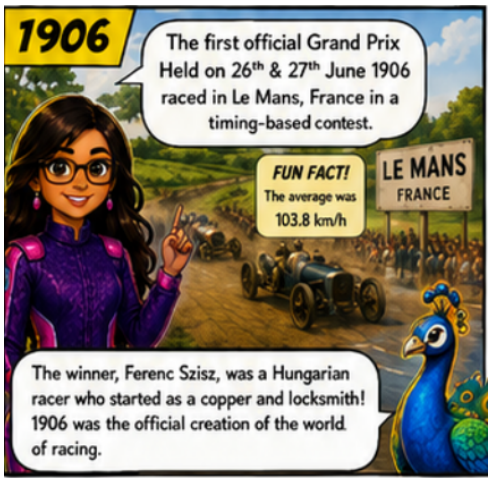
to the fastest heavy car. The rumbles of these cars on these very streets you are seeing are where our racing world was first created. As you can see, the flag was waved, and these early petrol vehicles rumbled their engines. These moments are what marked the creation of our world. **Did you know the Grand Prix's winner was Maurice Farman?** He was not only a racer but also an aviator, French aircraft designer, and manufacturer. The Grand Prix was a significant part of racing history.

1904

Now let us move forward in time. Past -O -meter voice, "Teleporting you to 20th June 1904 in France." Now come on, drivers ride into the meeting that laid the foundation for the creation of the FédérationInternationale de l' Automobile (FIA). Their headquarters are in Paris, France. This marked the start of efforts on the safety and management of the car racing sport. As we all know, safety is

crucial in our field because of the risk of accidents at the speeds we go. The world of racing was safer than it was before.

1906



Now, campers, let us move to 1906, when the first official Grand Prix was held, since the one held in 1901 was a localised unofficial race. Did you know that this Grand Prix was held around Le Mans, France, on a triangular circuit (**Fun Fact: It measured 103.8 km)?** The Grand Prix was held on 26th and 27th June 1906. Did you know that the winner, Ferenc Szisz, is a Hungarian racer who started as a copper and locksmith (The only copper he used later in life was in his cars!), but then the amazing power of automobiles fascinated him to be an automobile engineer, which led him to the world of racing!

Parts Of The Early 1900s Cars:

This is to explore the parts of the old car. As they say, old is gold; let's inspire ourselves with this and have fun. So let us ride in. We will

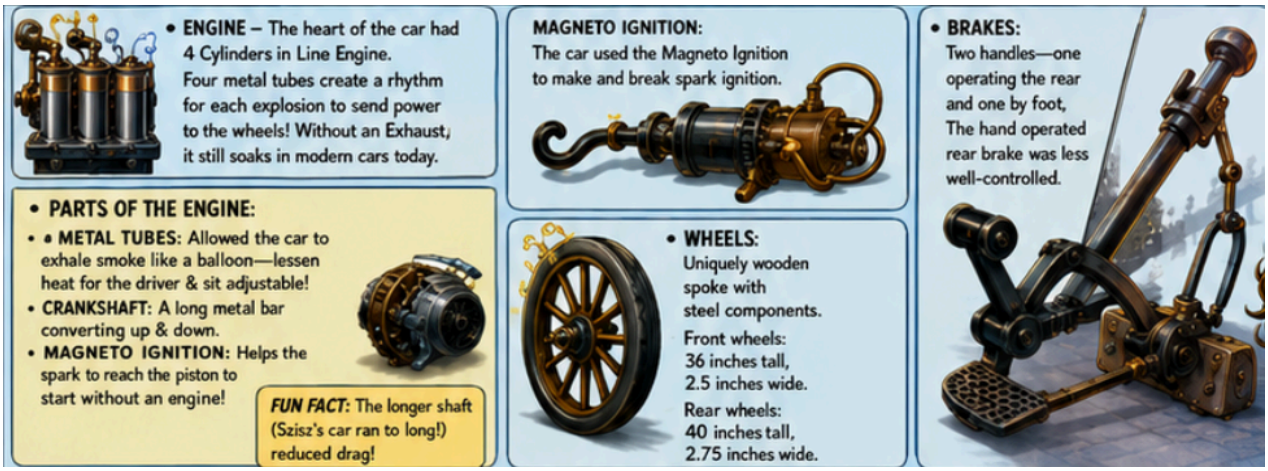
explore this using the Mercedes 35hp **(Fun Fact: Known as the first-ever modern car)**

1. Engine - The heart of the car had 4 Cylinders In Line Engine This means the engine is not only made of metal tubes, which causes the car to vibrate heavily and then pause, but it is made up of four metal tubes, which causes the vibration to go in a rhythm, making the car go faster. **Did you know that this framework is present as the basic structure of present-day cars?**

Parts of the Engine:

- **4 cylinders:** allowed the car to reach speeds like 80km/hr which was super fast for that time. This allowed a 5.9 L displacement.
- **Crankcase:** A long metal bar housing around the Crankshaft.
- **Honeycomb Radiator:** Bees living in the engine! No, it is the inner cooler of the car. When the engine heats up, there is water near the engine to cool it down. Then the water is sent to the honeycomb radiator to cool it so that it doesn't evaporate. (**Fun Fact: The HoneyComb Radiator was made by Wilhelm Maybach**.)
- **Magneto Ignition**: The car used the Magneto Ignitor to make and break spark ignition.

2. **Wheels:** Usually ,the wheels were made with wooden spokes and steel components. **Did you know that the front wheels were 36 inches tall and 3.5 inches wide, and the rear wheels were 40 inches tall and 4.7 inches wide!**
3. **Brakes:** There were two brakes, one operated by hand and the other by foot. But the hand-operated main shaft was water-cooled.



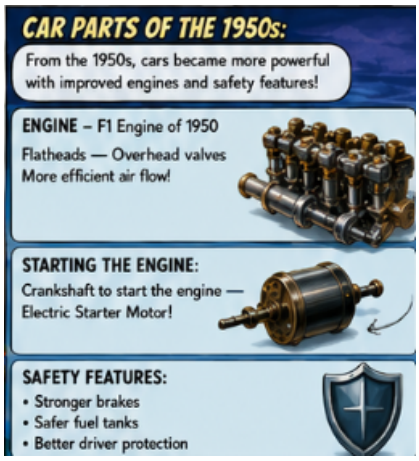
1950

Past-O-Meter, "Transporting you to the championship of 1950." The date May 13th, 1950, was the year in which the first-ever international championship race was conducted, and the F1 World Championship was created with a boom. Did you know the race was held at the Silverstone circuit (Fun Fact: The distance is 4.649 km and there were 70 laps) in Northamptonshire, UK! Until now, it was just car racing history; now it is F1 history. *Did you know that the first champion was Giuseppe "Nino" Farina, who, when aged nine, started to ride a two-cylinder "Temperino"!* Amulya particularly likes this era because this is the start of F1, and she loves the F1 sport. The championship included 6 major European Grand Prix and one American Grand Prix (Britain, Monaco, Switzerland, Belgium, France, Italy, Indianapolis 500)

1958

Past-O-Meter, " Transporting you to 1958." This year, the championship title was won by Mike Hawthorn. This was also the year F1 lost phenomenal racers Luigi Musso, Peter Collin,d Stuart Lewis-Evans, and Pat -O - Connor . This is a sad bit of F1! The year was tragic, but the season was an interesting one. The year was the bridge between the old and modern eras. **Did you know that the first female racer, Maria Teresa de Filippis, started her Formula 1 career in 1958?**

Car Parts of the 1950s



Car Parts of the 1950s: Since we understood the 1900s car,s let us see the powerful changes in the car from the engine to safety features.

Engine: From the 1900s, the car became more powerful, and the engine designers were super confident in their design. Let us see the parts of the F1 Engine of 1950 that have evolved:

Flatheads - Overhead valves: The flatheads used in the 1900s were very inefficient for their function. The Overhead valves were more efficient for air to flow directly through the cylinders.

Starting the Engine: In the 1900s, there was a crankshaft to start the car, which was exhausting and dangerous for the driver, whereas in the 1950s, the electric starter was less dangerous and less exhausting.

Cooling the water: (Ice Cooling time !) In the 1900s as you read in the previous pages, they used the honeycomb radiator which was made of simple copper tubes. Whereas the 1950 used Dry Sump Oil Systems helped in increasing horsepower and better lubrication and Closed Cooling System, which lubricated and cooled the water .

Supercharger: In the 1900s, cars sucked any air around there whereas the 1950s used mechanical pumps, which doubled or tripled the power generated.

Suspension: An independent form of suspension was used. Transverse leaf springs **(not the actual leaf but its one long strip of metal running through the axle accessing suspension)** or coil springs were used.

Steering: Rack and Pinion System. This system converts one kind of motion to another using a circular object (Pinion) and a long, toothed bar (Rack) . **(I kind of like the way it sounds, rack and pinion)**

Brakes: Drum Brakes **(Fun Fact: It is known as the drum brakes because of its hollow and cylindrical shape)** were mostly used, but it was a refined version.

Tires: Extremely narrow tires with High Aspect ratios (it means it is way taller than it is wide) placed on wire-spoked wheels.

1960 - Golden Era Of Formula 1!

Past- O - Meter, "Transporting you to the 1960s, the golden era!" The era lasting from the 1960s to the 1970s was very glamorous yet very tragic. The Lotus team won their first Formula 1 trophy with Stirling Moss in Monaco in 1960. With the emergence of Aerodynamics by the Lotus cars with

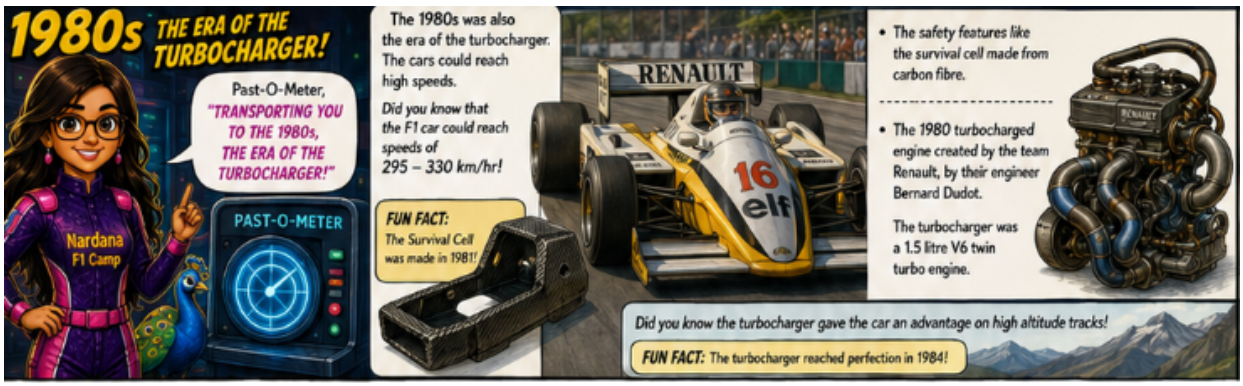
ground effect optimization, the creation of the vacuum, all of this started out with the Lotus 78 and later advanced to the 79. The era was tragic. We lost 14 drivers in the 1960s and 12 in the 1970s. Did you know, in this era, the safety of the F1 driver advanced from not wearing seatbelts to fireproof suits,

helmets, and harnesses! (Fun Fact: the first turbocharged car was created by Renault in 1977)

1980s - The era of the turbocharger

Past- O - Meter, "Transporting you to the 1980s, the era of the turbocharger." The 1980s were also the era of the turbocharger. The cars could reach high speeds.

Did you know that the F1 car could reach speeds of 295 - 330km/hr ! Safety features like the survival cell made from carbon fibre were developed in this era. **(Fun Fact: The Survival Cell was made in 1981)** The 1980 turbocharged engine was created by the Renault team, by their engineer Bernard Dudot. The turbocharger was a 1.5-litre V6 twin-turbo engine. Did you know the turbocharger gave the car an advantage on high-altitude tracks? (Fun Fact: The turbocharger was made advanced in 1984.



1990s - The era of the technological embark

Past - O - Meter," Transporting you to the 1990s, the era of the technological embark" The 1990s seasons included the tragic deaths of Ayrton Senna and Roland Ratzenberger, the rise of the great driver Michael Schumacher. The Driver Aids were banned very soon because of the risk of accidents, the reduction of drivers' skills during races, and their high costs caused the less funded teams difficulties in managing their funds. **Did you know that the rule was banned in the year 1994, when the first invention of the Driver Aids started in the 1990s!** The 1990s also advanced the safety of the sport. In 1994, the invention of a wooden skid block to provide more grip for the car provided advantages. Also, there were pit lane speed limits introduced for safety. Lastly, the circuits were redesigned to prevent any accidents.

3 major inventors in the F1 industry

So we have explored all the developments of the car and the sport. But now let's learn something unique. Inventors are very

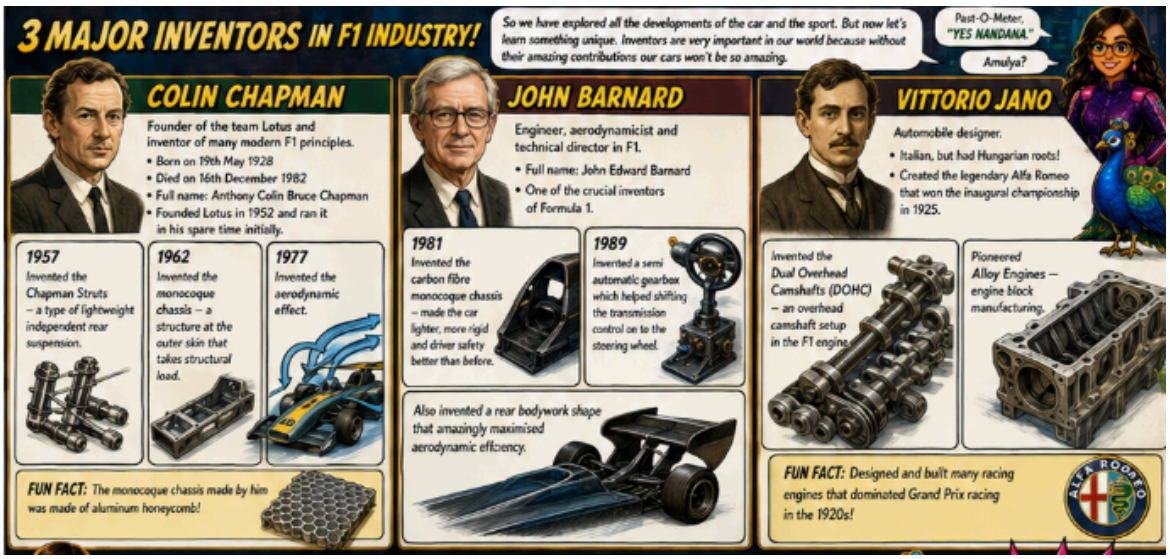
important in our world because without their amazing contributions, our cars wouldn't be so amazing. Ready Past - O - Meter! Past-O-Meter,"Yes, Nandana." Amulya? Amulya,"My favourite part, always ready!" Then let us explore the major inventors of F1!

I. Colin Chapman: The founder of the team Lotus and inventor of many modern F1 principles. He was born on 19th May 1928 and died on 16th December 1982. *Did you know that his full name was Anthony Colin Bruce Chapman?* He founded Lotus in 1952 and ran it in his spare time initially. Under his leadership pioneered the Chapman struts (It is a type of lightweight independent rear suspension) were pioneered as rear suspension devices in 1957. A little later, he invented the monocoque chassis in 1962 (a structure at the outer skin that takes structural load). *Fun fact: The monocoque chassis made by him was made of aluminum honeycomb.* He even invented the aerodynamic effect in 1977.

II. John Barnard : He was an engineer , aerodynamicist and a technical director in the field of Formula 1. *Did you know that his full name was John Edward Barnard!* He was one of the crucial inventors of Formula 1 . He was the father of the Carbon Fibre monocoque chassis. He invented this in 1981 . The material helped in making the car lighter , more rigid and made driver safety better than before. He made a semi automatic gearbox invented in 1989 , which helped shifting the transmission control on to the steering wheel. He even

invented a rear bodywork shape that amazingly maximised aerodynamic efficiency .

III. Vittorio Jano : He is an automobile designer. *Did you know that he was Italian but had Hungarian roots!* He created the legendary Alfa Romeo that won the inaugural championship in 1925 . He invented the Dual



Overhead Camshafts (DOHC) , an overhead camera setup in the Formula One car. He also pioneered Alloy Engines engine block manufacturing.

Jokes

1. Why did the 1980 car time travel to 1990?

➔ Because it wanted an upgrade!

2. What did the supercharger tell the 1901 engine?

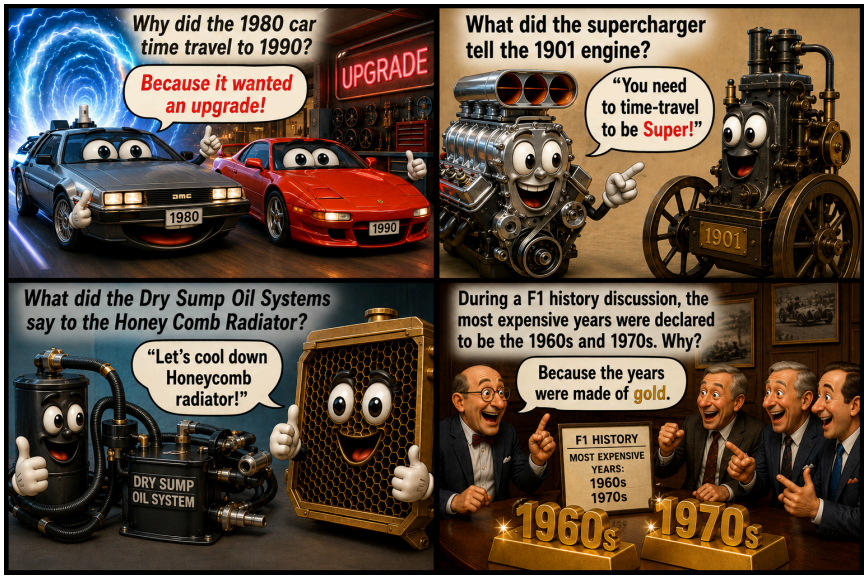
➔ " You need to time -travel to be Super!"

3. What did the Dry Sump Oil Systems say to the Honeycomb Radiator?

➔ "Let's chill Honeycomb radiator!"

4. During an F1 history discussion, the most expensive years were declared to be the 1960s and 1970s. Why?

➔ Because the years were made of gold.



Quiz

1. What was the advantage of the turbocharger?

2. When did the Lotus team win their first trophy?

3. When did F1 racing begin?

4. When was the FIA created, and what is the full form

5. When did Driver Aids get banned?

6. When was the turbocharger created?

FUTURE MECHANICS

Hello future champions, I'm Jaskirat — a pro, prestigious and prime F1 driver of the Neon Raptor Vortex GT-X with my pet, Igneous the lightning, shape-shifting iguana, so he can turn into almost everything, including different types of iguanas! Remember that now! And welcome to my

first chapter in Formula Racing Camp. Here, we'll talk about aerodynamics in F1, wings, and how they affect awesome races!

Aerodynamics

First of all, what is aerodynamics - the air doing gymnastics? Umm, no. Aerodynamics is the study of how air moves around objects

and how that movement affects them. In simple words:

It explains how air pushes, pulls, or flows around things that move, like cars, bikes, aeroplanes, or even a football. Aerodynamics is how air decides to be annoying.

So we make cars, bikes, and planes **smooth and pointy** so air goes: "Oh, okay, I'll just slide past!"

If air hits you hard → you go slow.

If air slips nicely around you → *Speed!*

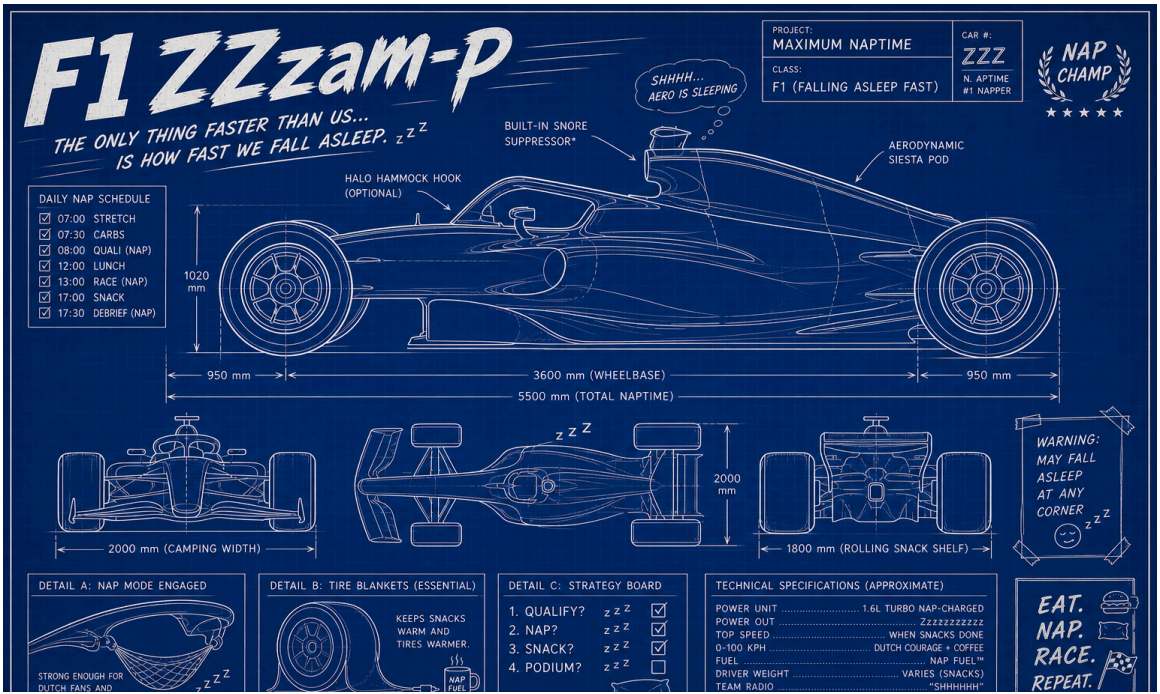
It's the car's invisible superpower, controlling air to make it faster, stickier in corners, and more stable at insane speeds. The wings push the car down, creating downforce—so strong that, in theory, an F1 car could drive upside down in a tunnel! Without aerodynamics, cars would be like kites at 300 km/h, sliding and struggling to stay on

track. Here's a joke: Why don't F1 cars ever get lost? Because they always follow the downforce!

If aerodynamics didn't exist, F1 wouldn't really exist & so wouldn't this super cool F1 book (That is BAD!!). Here's why: Formula 1 cars rely on aerodynamics to generate downforce, which pushes the car onto the track and gives it grip.

Without it, cars would slide around in corners, struggle to stay stable at high speeds, and could even lift off slightly at over 300 km/h. Races would be

Slower, far less precise, and much more dangerous—so the sport would be completely different, more like high-speed racing than the extreme engineering spectacle we see today. **Joke Time:** Scientists & coaches are still investigating why this wingless F1 car makes drivers instantly sleepy. Some blame the smooth aerodynamics; some blame the ultra-comfy cockpit… but after 3



laps, every test driver started snoring. That's why we proudly call it the **F1 ZZZAM-P!** — the fastest machine from 0 to record time!!

Wings

The wings on an F1 car are used to control the air so the car can go faster without sliding off the track. When an F1 car moves, air flows over and under its wings. The wings are shaped in such a way that air pushes the car down toward the road instead of lifting it. This pushing-down force is called downforce. Downforce makes the tyres grip the road better, especially when the car is turning at very high speeds. Without wings, an F1 car would skid, slide, or even lose control in corners. So even though the wings look cool, their real job is to help the car stay stable, fast, and

safe. ***Oh, great, Igneous made a toy line figure by shape-shifting into one! But it shows you where the different types of wings an F1 car has!***

Why F1 wings are important:

- **More grip** – The car sticks to the road like a magnet
- **Better cornering** – Cars can turn fast without slipping
- **Higher speed** – Drivers can go faster safely
- **More control** – The car stays balanced during braking and turning

So, F1 wings make the car *hug the road*, not fly in the air!

The History of Aerodynamics in F1:

1. 1950s - 1960s: Cars had no wings, just sleek bodies to reduce drag. Drivers relied on engine power and skill. And Igneous hates it!

2. Late 1960s: First wings appeared to push cars down for more grip in corners. Early high wings were unstable and risky.

3. 1970s: Ground effect introduced—shaping the car's underside to suck it onto the track for faster cornering.

4. 1980s–1990s: Wings and bodywork became more refined; wind tunnels and computers used to optimize aerodynamics.

5. 2000s: Cars had complex wings and diffusers; airflow management became key for speed and stability.

6. 2022–Today: Modern F1 cars balance downforce and drag, with ground-effect floors returning to allow closer racing and more overtakes.

Today's Formula 1 cars are not just machines—they are masterpieces of airflow engineering. Every curve, wing, and surface is designed to control invisible streams of air with extreme precision. Teams use advanced simulations and powerful computers to test thousands of designs before a car even hits the track. The return of ground effect has transformed racing, letting drivers chase each other wheel-to-wheel at incredible

speeds. In this era, victory doesn't just belong to the fastest engine—it belongs to the team that can command the air itself!

The Cost of an F1 Car's Wings

F1 cars' wings are very expensive because they are made from carbon fibre, designed in wind tunnels, and tested again and again.

- Front wing: roughly $150,000–$300,000
- Rear wing: roughly $100,000–$250,000
- With development and testing, a full wing design can cost millions over a season.

That's because even tiny shape changes can affect speed, grip, and lap time.

My car's wing, which was built in the UK and upgraded in Maranello, Italy, has my name built into the carbon itself, and when the car powers up, electric & flame energy ignite along the bottom of the wing—for looks, speed, and a warning to thieves. It has advanced aerodynamics – still produces strong downforce while hiding the tech inside. The shock system is for super-fast speed, and with Igneous, it powers up a lot more!

Wheels of F1 Cars (Slick tyres)

Ah yes! I almost forgot about the **smooth, cartoon-like wheels of F1 cars**—also called **slick tyres**—one of the most recognisable features on track, and a **huge reason** why F1 cars are insanely fast.

Unlike regular road tyres, F1 tyres have no tread patterns, which makes them perfectly smooth. You might wonder, "Won't that make them slippery?" Actually… It's the opposite! Good! I thought Igneous would think cartoons are more important than them!

Slick tyres are designed to maximise contact with the track, giving the car the most grip possible. More rubber touching the asphalt = more friction = faster cornering and braking.

Which means drivers can push harder through turns without losing control.

F1 tyres are engineered to handle crazy speeds and temperatures—up to 100°C or more! That's hotter than a frying pan! (And yes… no idea how frying pans entered the chat either).

Now imagine driving an F1 car on a rainy day with slick tyres. Do you think the car would grip the road well, or slide everywhere like it's on soap?

(Hint: slick tires love the dry tracks, not puddles !)

That's why F1 teams choose different tyres for different conditions —soft, medium, hard, and wet—to balance speed and durability, and drivers are trained to handle almost all types.

So those smooth, "cartoon" tyres aren't just for looks —they're high-tech performance machines. Okay… maybe they are *also for* looks, but mostly it's about performance!

If F1 Tires Were Normal Tires

But if the cool, unique F1 tyres were replaced with normal road car tyres, the car would be almost impossible to drive!!

See, F1 cars are super light, extremely fast, and have huge wings that push the car hard onto the road. Normal tyres are not made to handle that much speed, force, and heat. As soon as the car tried to go fast or turn, the tyres would lose grip, slide, or even burst. The car wouldn't be able to use its aerodynamics properly because the tyres wouldn't "hold on" to the road.

This is what would go wrong:

- **Very little grip** – normal tyres would slip and slide
- **Overheating** – they'd melt or wear out very fast
- **No fast cornering** – the car would spin or crash
- **Poor braking** – stopping would take much longer
- **Possible tyre failure** – at F1 speeds, they could burst!

The Engine (Heart of an F1 Car)

Alright, campers… We've already explored the wings that push the car down and the tyres that grip the track. But now, it's time to uncover the real beast hiding inside the machine… the engine — or as F1 calls it, the *power unit*.

The power unit provides the energy that helps the car accelerate and perform on track. In the simplest way possible, it's the heart of the car.

- **No engine = no speed**
- **No speed = no race**

And no race? Yeah… that's a disaster!

But F1 Engines Are NOT Normal

Here's where things get crazy. F1 engines are nothing like the ones in normal road cars. These machines use a super-advanced hybrid power unit, meaning they use both fuel and electricity together.

That combination creates insane power that pushes the car to extreme limits.

How Powerful Is It?

So how powerful are we talking? Modern F1 power units produce around 1000 horsepower, depending on the car and conditions. Not actual horses—but the force of multiple cars combined into one machine.

- **Speeds over 350 km/h**

That's what makes an F1 car feel like a rocket on wheels.

Special Features

What makes it even cooler is how smart this system is. F1 cars use an Energy Recovery System (ERS), which captures energy during braking and from exhaust heat, then reuses it to provide extra power, like a hidden boost button.

They also use a turbocharger, which compresses more air into the engine so more fuel can burn efficiently and produce more power.

Why the Engine is Important

The engine is one of the most important parts of the car.

- **Raw speed**
- **Acceleration**
- **Overtaking power**

It works together with aerodynamics to push performance to the limit.

Cost of an F1 Engine

Now for the shocking part… the price. An F1 power unit costs around $10–15 million (₹80–120 crore).

That's because it's not just an engine—it's a hybrid masterpiece built with cutting-edge technology, advanced turbo systems, energy recovery tech, and materials that can handle extreme conditions. Even the smallest upgrade can change the outcome of a race.

Famous Racers & Their Unique Engines

Over the years, legendary drivers have used some of the most powerful and iconic engines in F1 history.

Lewis Hamilton drove cars powered by Mercedes hybrid engines, known for their incredible efficiency and smooth power delivery. These engines helped him dominate the turbo-hybrid era and win multiple championships.

Michael Schumacher became a legend with Ferrari V10 engines in the early 2000s. These engines were famous for their screaming sound and raw power, helping Ferrari dominate F1 during that time.

Ayrton Senna used the iconic Honda turbo engines in the late 1980s. These engines were extremely powerful and aggressive, perfectly matching his fearless driving style.

Max Verstappen currently uses Red Bull Ford Powertrains (Honda-based hybrid engines). These engines are known for their explosive acceleration and strong performance, helping him dominate modern F1 races.

F1 engines aren't just machines…
They're legend-makers!

Different engines + different drivers = different eras of domination

My Engine

The Neon Raptor Vortex GT-X doesn't just have AN engine…

It has a **Quantum Storm Hyper-Drive Core!**

This isn't some normal hybrid system.
This is a fusion of **lightning, fire, and pure racing energy**.

What Powers It?

- **Lightning Core Reactor** – Generates electric bursts faster than a thunderstrike!
- **Igneous Flame Sync** – My iguana literally powers the engine with living fire energy while standing on the wing!
- **Turbo Cyclone Compressors** – Spin air into a mini storm for insane acceleration!

What Happens When It Starts…

When I press the ignition:

The engine doesn't just "start"...

IT AWAKENS

- The car glows with electric energy
- Flames ripple across the rear vents
- The sound? Not a normal roar…

It sounds like a **thunderstorm ripping through a volcano!** *(Weird description but okay…)*

Performance Mode

When I activate full power:

- Energy surges through the chassis
- Flames boost acceleration
- Air bends around the car

The car doesn't just go fast…

It TELEPORTS forward like a lightning strike!

Jokes

Ok, campers… It's joke time! Igneous & I made some windy, rolly & growly jokes here, perfect for this chapter.

- **Why did the aerodynamicist get kicked out of the library?** He kept trying to adjust the "drag" on the floor with the books!

- **How do F1 engineers stay cool?** They have a lot of fans… and they're all in the wind tunnel!

- **Why was the track so stuffy?** It's not 'cause the crowd left, well, maybe, but it's because there are messy tyres, rusty engines & everything lying on the tracks! Igneous, don't spread hot air everywhere!

- **Why did the tyre cross the road?** It didn't. Why do you ask? He didn't want to!

- **Why did the tyre go to therapy?** Because it couldn't handle the pressure!

- **What does the last F1 driver of the race like to eat?** The dust of the others' tyres!

- **What's an F1 car's tyres' favourite type of music?** Rock n' Roll!

- **What did the tyre say after the race?** He said he was very tired! But Igneous is 100000+ times more tired as a tire than a normal tire!

- **Why are F1 tires so bad at keeping secrets?** Because they always leak information through telemetry!

- **Why was the F1 engine so confident?**
  Because it had *a lot of horsepower behind it!*

- **What did the engine say before the race started?**
  "Let's make this a *rev*-olution!"

- **Why don't F1 engines ever get tired?**
  Because they're always *revved up* and ready to go!

Chapter 1 Quiz!

Multiple Choice

1. What is aerodynamics?
A) The study of engines
B) The study of how air moves around objects
C) The study of tyres
D) The study of fuel

2. Downforce helps an F1 car...
A) Fly into the air
B) Stick to the road
C) Use less fuel
D) Change tyres

3. Which part mainly creates downforce?
A) Steering wheel
B) Wings
C) Mirrors
D) Seat

4. Why are F1 tyres smooth?
A) They are cheaper
B) They create more grip on dry tracks
C) They look better
D) They reduce weight

5. Which tyre is best for heavy rain?
A) Slick
B) Soft

C) Wet tyre
D) Medium

6. Modern F1 power units use...
A) Steam
B) Petrol only
C) Fuel and electricity
D) Solar panels

7. ERS stands for...
A) Engine Racing Speed
B) Energy Recovery System
C) Engine Rotation System
D) Electric Racing Setup

8. Approximately how much horsepower does a modern F1 car produce?
A) 300 hp
B) 500 hp
C) 700 hp
D) Around 1000 hp

9. Which material are F1 wings mainly made from?
A) Steel
B) Aluminium
C) Carbon fibre
D) Plastic

10. Ground effect mainly works by...
A) Cooling the engine
B) Sucking the car towards the track
C) Reducing fuel use
D) Increasing tyre size

True or False

11. Aerodynamics only matters for aeroplanes.
12. More downforce usually gives better cornering.
13. Slick tyres work best in heavy rain.
14. Modern F1 cars use hybrid power units.
15. Without wings, F1 cars would be harder to control.
16. Engines and aerodynamics work together.
17. Formula 1 began using wings in the late 1960s.
18. Carbon fibre is heavier than steel.
19. Wind tunnels help engineers design wings.
20. An F1 engine costs millions of dollars.

Fill in the Blanks

21. _____ is the force that pushes the car onto the track.
22. Smooth F1 tires are called _____ tires.
23. F1 wings are mainly made from _____ fibre.
24. The _____ stores and reuses braking energy.
25. Modern F1 cars can reach speeds above _____ km/h.
26. The engine is often called the _____ of the car.
27. Ground effect helps increase _____.
28. More rubber touching the road creates more _____.
29. Air flowing around the car affects its _____.
30. Formula 1 first started in _____.

MAP MASTERS

Formula racing is always known for being the fastest circuit sport in the world. What few people know about F1 racing is that it is among the geographically diverse sports ever created. In every competition, the location is completely new and unique, and it changes everything from how the car runs to how the drivers drive and even how the races are conducted.

Every race is not only against the competition; it's also against nature. Everything, from the ground the car drives on to the air, the weather, and even the streets of the city, all contribute to how the game turns out.

We have gone through history, into the future, and seen the now.

Today we go deeper into… LOCATION

And in Formula 1 racing… the world itself becomes your racetrack.

Seas & Mountains: Monaco

Welcome to Monte Carlo, the place where the famous Monaco Grand Prix takes place.

Monaco can be called a unique destination in Formula 1 racing since this destination is located in between steep hills and the beautiful Mediterranean sea coast. The Monaco track is not a track created specifically for races; it is made of the streets of the town itself, which means that there cannot be any mistakes while competing here.

Driving around this circuit requires a lot of skill due to small roads, difficult turns, and lots of changes in elevation, but we shouldn't forget about challenges such as bumpy driving surfaces, building shadows, and fast changing illumination (when leaving a dark tunnel).

The Fairmont Hairpin is one of the slowest turns in Formula 1, and still The precision in maneuvering is essential while going through it.

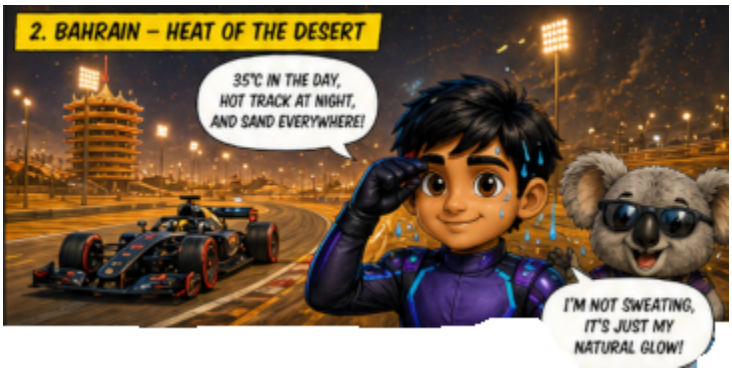
This location is special not only due to the difficulty of the circuit but also the surrounding atmosphere as well.

The Bahrain International Circuit

The Bahrain International Circuit combines the elements of the desert into the race itself.

This circuit is based in the Middle East and finds itself surrounded by the dry and sandy land that is characterized by hot temperatures during the daytime. The racing event takes place at night time, but this does not make the surface cool since the accumulated heat stays on the asphalt.

One of the aspects that should be taken into account while driving through such conditions is that the asphalt becomes heated due to the sun, which makes tires prone to overheating and losing grip.



Moreover, another important element is that sand can blow into the racing track, affecting the braking distance and grip.

Dry air affects the engine and cooling system, and therefore careful calculations should be made by teams.

Circuit de Spa-Francorchamps

Welcome to Circuit de Spa-Francorchamps , one of the most legendary tracks on the planet!

The circuit is notorious for its long distance, varying elevations, but above all else, for the capricious nature of the weather conditions there.

Due to being surrounded by forests and hills, the track is subjected to its unique weather phenomenon.

It is quite possible that one part of the circuit will be bone-dry while others might be under heavy rainfall. This means that the driver is required to keep adjusting his or her approach during the race.

Perhaps the most well-known turn here is the Eau Rouge and Raidillon turn, where the car ascends steeply up the hill.

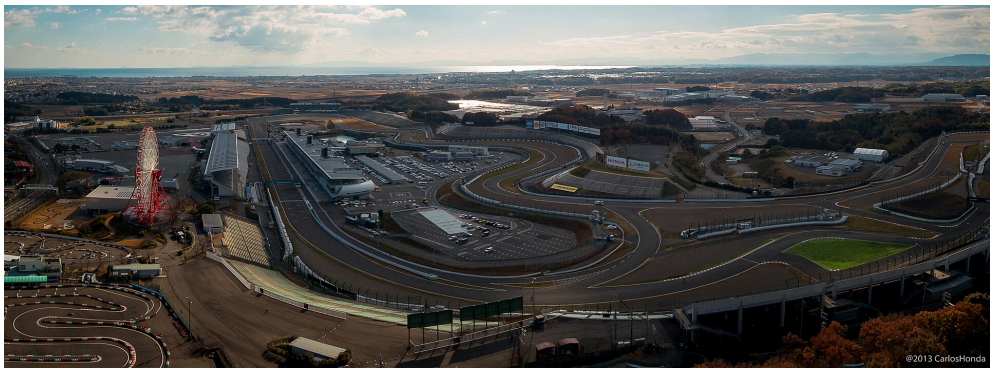
Spa teaches every driver a valuable lesson:

Be prepared for anything!

Suzuka International Racing Course

Racing at the Suzuka International Racing Course 🇯🇵 is a true art of accuracy.

Suzuka is one of the rare circuits that form a figure-eight shape, as one part of the circuit passes over another using a bridge. Such a design gives rise to a rhythm that racers need to understand.



"Turns of S" require smooth driving and impeccable timing. Every step, whether braking, accelerating, or handling the wheel, must be calculated precisely.

Consistency and accuracy will get you through here. While street races offer speed, Suzuka gives it to those who remain balanced.

Passionate and respectful Japanese fans are what make the race even more exciting.

Suzuka is a track where:

accuracy is key.

Autodromo Nazionale Monza

Welcome to Autodromo Nazionale Monza 🇮🇹, also called the Temple of Speed.

Monza is one of the oldest race tracks in F1 and is well-known for its long straight sections and fast-paced racing.

Due to this track's long straight portions, slipstreaming becomes crucial for drivers. They try to drive close to another vehicle to minimize drag and increase their speed.

Nevertheless, due to its fast pace, the circuit has numerous braking sections. The drivers need to brake rapidly when approaching chicanes.

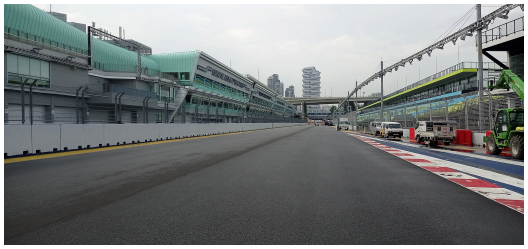
The energy from the Tifosi makes the atmosphere unique.

All efforts made by the drivers at Monza aim for one goal:

maximizing speed.

Marina Bay Street Circuit

The Marina Bay Street Circuit 🇸🇬 takes place during the night.



It was the first Formula 1 night race and continues to be extremely challenging. The humidity and temperature make it physically demanding on drivers.

Drivers can lose up to 3 kg of their body weight as a result of sweating. It is a long race with a lot of corners that requires concentration.

The circuit itself is narrow and rough, making it difficult for racers.

The bright lighting makes reflections off buildings more visible.

Singapore pushes humans to their physical limits.

Autódromo Hermanos Rodríguez

Altitude matters at Autódromo Hermanos Rodríguez 🇲🇽.

With an elevation of more than 2200 meters, there's less oxygen in the air.

This results in lower engine power output and less aerodynamic downforce.

Circuits are slippery, braking is challenging, and there's less traction available.

Teams need to adapt their settings to account for low air density.

Fans in the stands in the stadium sector create an electrifying ambiance.

Here, geography influences even the intangible:

the air.

Albert Park Circuit

Race Dynamics at Albert Park Circuit 🇦🇺

Due to its proximity to the sea, the race track in question is affected by the dynamic weather conditions. Wind gusts influence car handling, particularly around fast corners.

Cooler temperatures make handling tires more difficult.

As the race weekend progresses and the number of cars on track increases, grip improves.

Drivers have to be aware of nature in Australia.

Nature cannot be seen—but it is always there.

Las Vegas Strip Circuit

At the Las Vegas Strip Circuit 🇺🇸, Formula 1 enters a new era.

Bright neon lights, long straights, and a city built for entertainment create a race like no other.the race here are less.

Buddh International Circuit

Welcome to the Buddh International Circuit 🇮🇳

Formula One in India is a thrilling combination of speed, tradition, and excitement. Situated in close proximity to the capital of India, New Delhi, the circuit is characterized by a wide design, high-speed sections, and smooth corners.

It is equipped with an extremely long straightaway at which cars go at exceptionally high velocities, followed by aggressive braking zones. Drivers need to quickly switch from driving at full throttle to taking tight turns.

As far as the climate is concerned, India boasts a warm and dry climate that negatively impacts tire traction and causes accelerated tire wear. Similar to Bahrain, tire management is crucial.

What is more, there is one more feature that makes racing in India unique – the fans' enthusiasm. They scream enthusiastically and cheer every driver's accomplishment. You feel the power of the nation on your shoulders.

Experience:

The drive feels exhilarating and invigorating.

The track flows effortlessly, but requires utmost attention.

The heat challenges you, and the crowd inspires you.

This is where racing feels like home.

Koala:

Fast cars AND amazing food? I'm never leaving!

Travelling from one circuit to another shows how Formula 1 is geographically diverse. The world of Formula 1 is not only speed and the car but also the surroundings. Like how circuits in Monaco are different from the ones in Suzuki. The climates, the race tracks, the various hairpin bends, traditions, and culture. This is me, Tejas, signing off…… Wait, wait, let's finish a Fun quiz and jokes at the end.

Quiz Section

1. Why is location important in Formula 1?
2. In which country is Monte Carlo located?
3. Which circuit is called the "Temple of Speed"?
4. What type of race is Singapore known for?
5. Which track has a figure-8 layout?
6. What makes Spa's weather unique?

Jokes

1. **Why don't F1 drivers get lost?**
   Because they always follow the racing line! 🏁
2. **Why is the Monaco Grand Prix so hard?**
   Because even Google Maps says: "Make NO mistakes!"
3. **Why did the car struggle at the Bahrain International Circuit?**
   Because it couldn't handle the *sand-wich* conditions!
4. **What did the driver say at Circuit de Spa-Francorchamps?**
   "Do I need wet tires… or sunglasses?"

5. **Why do drivers love Autodromo Nazionale Monza?**
   Because life is better in the *fast lane!*
6. **Why was the driver so tired at the Marina Bay Street Circuit?**
   Because even the air said, "I'm heavy today!"